



Midterm Exam	Duration: 30 minutes	Total Marks: 10
Total Number of Questions: 1		

1st Question	Marks: 10
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- A) [5 Marks]** In SIFT feature detector and descriptor, explain the function of the following:
- Scale space
 - Difference of Gaussian (DoG)
 - Orientation assignment
 - Normalization of the window's rotation and scale
- B) [5 Marks]** Sketch a general architecture of the Siamese Network with triplet loss used for face verification. Briefly explain the training procedure in terms of input, the learning objective and the loss function used.

Final Exam	Duration: 2 hours	Total Marks: 65
Total Number of Questions: 4		

2nd Question	Marks: 18
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State whether each of the following statements is True or False.

- If we initialize the k-means clustering algorithm with the same number of clusters but different starting positions for the centers, the algorithm will always converge to the same solution.
- The Harris corner detector is stable under image rotation.
- A good action recognition CNN-based model is the one that relies on extracting temporal features only between video frames to define what kind of motion/action occurs.
- Instance segmentation is considered a hybrid technique between image classification and object detection methods.
- In the image classification methods where each image belongs to one class of objects, the Histogram of Oriented Gradients (HOG) is considered a reasonable feature extractor for recognizing objects in the underlying images.
- Visual Words help in encoding images by representing them as distribution/histogram of feature occurrences.
- The YOLO algorithm for object detection and localization can predict the overlapped object locations in images.
- In the U-Net architecture for semantic segmentation, the up-sampling operation is based mainly on deconvolution operation.
- Applying Eigenfaces technique (using PCA) and nearest neighbor method to build a classifier for recognizing faces will give reasonable performance if applied directly to the following sample images.



3rd Question**Marks: 17**

A) [13 Marks] You are given the following pair of images containing multiple objects that have undergone motion from one frame to the next. Assume there exists a color constancy between both frames as the pixel's color looks the same. Also assume that the motion of pixels is too small (pixel's shift is less than 1 pixel). Your task is to estimate the displacement (optical flow) between image frames.



- Explain an algorithm to solve the aforementioned task, stating any additional constraints that it may require.
- After finishing applying your model, the results were unsatisfying. It appears that your assumption of small motion was incorrect and the shift in pixels was large. An idea to resolve this challenge is to iteratively lower the resolution of the two frames using sub-sampling to decrease the motion between image pixels. Explain an alternative to your algorithm that adapts to this idea.

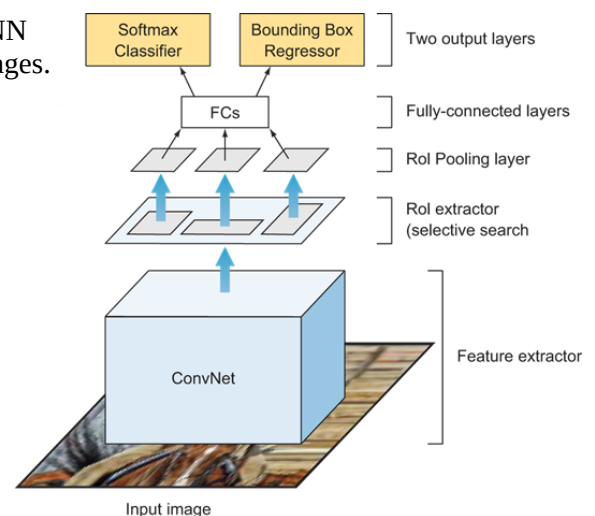
B) [4 Marks] You are working for a special effects company and want to have real actors inserted into a computer-generated world. One way to do this is to film the actors in front of a blue screen, so that the background can easily be removed and substituted with a computer generated one. Due to lighting variations, the background will not appear pure blue, so checking for a single intensity value might not work well. Instead, you decide to model the distribution of background colors as a probability function. You are given training images T with pixels labeled as foreground (F) and background (B).

Describe a model based on Bayes Rule that given a new image can be used to recognize any pixel to be either background or foreground.

4th Question**Marks: 17**

A) [6 Marks] The given diagram shows the Fast R-CNN network for object detection and localization in images.

- Explain the function of the ROI pooling layer.
- Sketch a modified diagram to represent the Faster R-CNN network.
- Explain (in points) the main changes between the two models.



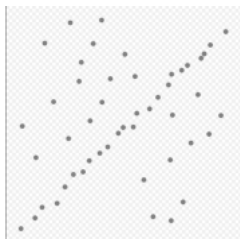
B) [11 Marks] Mean Shift is a popular method used in computer vision for problems such as image segmentation and object tracking.

- i. Describe the main steps of how Mean Shift is used to perform image segmentation, including the parameter(s) that must be set in order to use Mean Shift.
- ii. Comment on the sensitivity of the algorithm to its parameter value(s); i.e., what are the effects of setting the parameter(s) to values that are too big or too small.

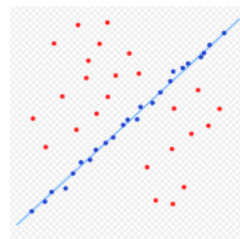
5th Question

Marks: 13

A) [7.5 Marks] Given the following 2-D point cloud, it is supposed to fit an optimal line model to it.



(a) Original point cloud



(b) Line model after fitting

- i. Describe how you will use **RANSAC** algorithm for the aforementioned task.
- ii. Assume that there are 60% inliers; compute the minimal number of RANSAC iterations needed to get, with probability 90%, at least one random sample that is free from outliers.

B) [5.5 Marks] In a Content Based Image Retrieval (CBIR) system, the purpose is to retrieve images from a large database using query by image content.

- i. Define the notion of the “semantic gap” in the context of systems for content based image retrieval.
- ii. Consider training two different ConvNets on a dataset with 10 different classes. After querying with the same cat image, the ranks of the retrieved images results as follows for the two models respectively:
Model 1: (cat , dog, dog, cat, cat, car,...)
Model 2: (cat, cat, cat, dog, car, car,...)
If the database contains only 3 cat images, use the Average Precision (AP) metric to evaluate both queries and determine which model is more accurate.

♥ With My Best Wishes ♥

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